

Pénélope Azuelos

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Education

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| 2022 – present | University of Bristol, UK
<i>PhD in Mathematics (ongoing), supervised by Mark Hagen and Indira Chatterji</i> |
| 2021 – 2022 | École Normale Supérieure de Lyon, France
<i>Master (M2) Mathématiques Avancées</i> |
| 2017 – 2021 | University of Bristol, UK
<i>Master in Science in Mathematics (with first class honours)</i> |

Publications

(with Damien Gaboriau) *Perfect kernel and dynamics: from Bass-Serre theory to hyperbolic groups*, Mathematische Annalen 391: 4733–4789, 2025.
<https://doi.org/10.1007/s00208-024-03038-w>
On subgroups with narrow Schreier graphs, Bulletin of the London Mathematical Society, 56(12): 3652–3668, 2024. <https://doi.org/10.1112/blms.13157>

Preprints

A guide to constructing free transitive actions on median spaces, arXiv:2507.22230.

Teaching

Teaching assistant at the University of Bristol for the following courses:

Spring 2024	Introduction to proofs and group theory MATH10010
Autumn 2023	Combinatorics MATH30030
Spring 2020	Introduction to geometry MATH20004

Talks

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| June 2025 | Conference on hyperbolicity and generalizations, HIT, Harbin, “A guide to constructing free transitive actions on median spaces”. |
| February 2025 | Bristol Junior Geometry Seminar, “Real trees and the groups that act on them”. |
| January 2025 | RSME’s 7th congress of young researchers, parallel session on geometric group theory, UPV/EHU, Bilbao, “Narrow graphs and virtual fiber subgroups”. |
| December 2024 | Séminaire d’Algèbres d’Opérateurs, IMJ-PRG, Paris, “Comment repérer des fibres virtuelles”. |
| October 2024 | Séminaire de l’équipe d’algèbre, topologie et géométrie, Laboratoire J.A. Dieudonné, Nice, “Comment trouver des fibres virtuelles”. |

September 2024	Séminaire de Théorie Ergodique, Institut de recherche mathématique de Rennes, “ <i>L’espace des sous-groupes : noyau parfait, transitivité topologique et actions moyennables.</i> ”
	– Algebra Geometry and Topology seminar, Heriot-Watt, “<i>Narrow graphs and virtual fiber subgroups</i>”. – World of Geometric Group Theory (online conference), “<i>A geometric criterion for detecting virtual fiber subgroups</i>”.
May 2024	International Young Seminar on Bounded Cohomology and Simplicial Volume, “ <i>Topology and dynamics on the space of subgroups</i> ”.
March 2024	Seminar: GdT Action ! ENS de Lyon, “ <i>Un critère géométrique pour déterminer si un sous-groupe est une fibre virtuelle</i> ”.
	– Conference: Rigidité en théorie des groupes : aspects géométriques et ergodiques, Université Paris-Saclay, “<i>Un critère géométrique pour déterminer si un sous-groupe est une fibre virtuelle</i>”.
February 2024	Cambridge Junior Geometry Seminar, “ <i>Ends of groups and related invariants</i> ”.
December 2023	Bristol Postgraduate Seminar, “ <i>Is it trivial? And other difficult problems</i> ”.
November 2023	Oxford Junior Topology and Group Theory Seminar, “ <i>Topology and dynamics on the space of subgroups</i> ”.
September 2023	Le séminaire virtuel francophone Groupes et Géométrie, “ <i>Topologie et dynamique de l’espace des sous-groupes</i> ”.
May 2023	Conference: Measured Group Theory, Stochastic Processes on Groups and Borel Combinatorics, CIRM Marseille, “ <i>Perfect kernels and dynamics: from Bass-Serre theory to hyperbolic groups</i> ”.
January 2023	Bristol Junior Geometry Seminar, “ <i>Subgroup spaces and actions on trees</i> ”.

Service

2025	Organisation of the Postgraduate Group Theory Conference 2025
2023 – 2024	Organisation of the Bristol Junior Geometry Seminar

Awards and funding

2026	Oberwolfach Leibniz Graduate Students travel grant to attend an Oberwolfach workshop.
2024	Grant from the Académie d’Excellence “Systèmes Complexes” for a three month research visit to the LJAD in Nice, supervised by Indira Chatterji.
2022 – 2026	University of Bristol PhD scholarship.
2021 – 2022	MiLyon scholarship for an M2 masters program.
2020	Summer research bursary funded by the Heilbronn Institute for Mathematical Research, supervised by Misha Rudnev.

Languages

English and French (native speaker)